



**DEPARTMENT OF THE AIR FORCE
HEADQUARTERS 96TH TEST WING (AFMC)
EGLIN AIR FORCE BASE FLORIDA**

4 December 2025

MEMORANDUM FOR: AFTC/PZIOA

FROM: 782 TS/RNWI

SUBJECT: Government Technical Requirement Document: SpIRS Hyper-spectral Imager brand name and an option to purchase one at a later date.

1. The 782nd Test Squadron (782 TS) Seekers and Sensors Test Flight (RNWI) has a requirement for a SWIR/MWIR Commercial Off the Shelf (COTS) **Spatially resolved InfraRed Spectrometer (SpIRS)**. The Hyper-spectral Imaging System (HIS) must meet the following technical requirements.
 - a. The spectrometer must have a spectral range to $1.5 - 5.5 \mu\text{m}$
 - b. The detector pitch must be $\leq 350 \mu\text{m}$
 - c. The spectrometer Focal Plane Array (FPA) must be $\geq 8 \times 8$
 - d. The spectrometer FPA must have zero dead pixels
 - e. The spectrometer must have variable aperture for signal attenuation
 - f. The spectrometer must have selectable spectral resolution 1 to $\geq 32 \text{ cm}^{-1}$
 - g. The spectrometer must have minimum scan rates at the following resolutions:
 - i. 3 Igrams/sec at 1 cm^{-1}
 - ii. 26 Igrams/sec at 8 cm^{-1}
 - iii. 44 Igrams/sec at 16 cm^{-1}
 - iv. 71 Igrams/sec at 32 cm^{-1}
 - h. The HIS dynamic range must be $\geq 10000:1$
 - i. The HIS must have a 10Gigabit Ethernet (GigE) output.
 - j. The HIS must include three (3) telescope assemblies
 - i. x0.5
 - ii. x2
 - iii. x6
 - k. The HIS must include a co-aligned bore sighting color video camera with manual varifocal lens with a focal length of 20 to 100mm (f#1.6)
 - l. The HIS must have a built-in GPS/IRIG B Timecode decoder interface.
 - m. The HIS must include an acquisition computer (Windows 11 or Linux) 64-bit architecture
 - n. The HIS must include software to acquire, analyze and display scientific data
 - o. The HIS operational voltage must be: 120V, 60Hz
 - p. The HIS enclosure must be hermetically sealed.
 - q. The HIS must come with manuals, installation and training at customer site.
2. The 782nd Test Squadron (782 TS) Seekers and Sensors Test Flight (RNWI) has an optional requirement for a MWIR/LWIR Commercial Off the Shelf (COTS) **Spatially resolved InfraRed Spectrometer (SpIRS)**. The optional Hyper-spectral Imaging System (HIS) must meet the following technical requirements.
 - a. The spectrometer must have a spectral range to $2.0 - 12 \mu\text{m}$
 - b. The detector pitch must be $\leq 350 \mu\text{m}$
 - c. The spectrometer Focal Plane Array (FPA) must be $\geq 8 \times 8$
 - d. The spectrometer FPA must have zero dead pixels
 - e. The spectrometer must have variable aperture for signal attenuation

- f. The spectrometer must have selectable spectral resolution 1 to $\geq 32 \text{ cm}^{-1}$
- g. The spectrometer must have minimum scan rates at the following resolutions:
 - i. 3 Igrams/sec at 1 cm^{-1}
 - ii. 26 Igrams/sec at 8 cm^{-1}
 - iii. 44 Igrams/sec at 16 cm^{-1}
 - iv. 71 Igrams/sec at 32 cm^{-1}
- h. The HIS dynamic range must be $\geq 10000:1$
- i. The HIS must have a 10Gigabit Ethernet (GigE) output.
- j. The HIS must include three (3) telescope assemblies
 - i. x0.5
 - ii. x2
 - iii. x6
- k. The HIS must include a co-aligned bore sighting color video camera with manual varifocal lens with a focal length of 20 to 100mm (f#1.6)
- l. The HIS must have a built-in GPS/IRIG B Timecode decoder interface.
- m. The HIS must include an acquisition computer (Windows 11 or Linux) 64-bit architecture
- n. The HIS must include software to acquire, analyze and display scientific data
- o. The HIS operational voltage must be: 120V, 60Hz
- p. The HIS enclosure must be hermetically sealed.
- q. The HIS must include user manuals.

I certify that the technical requirements contained in this document are accurate:

JASON MCDONALD
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